WORKSHEET 2.1

Slope at a Single Point

Unit 2: The Derivative

Key ideas

- The **average rate of change** of f on $[a, a+h]$ is the slope of a secant line: $\dfrac{f(a+h)-f(a)}{h}$.
- The **instantaneous rate of change** at $x=a$ is the limit of those slopes as $h \to 0$. It is the slope of the tangent line.
- The tangent line at $x=a$ passes through $(a, f(a))$: $y - f(a) = m(x-a)$.
- A rate of change has units of "output per unit of input".

Worked examples

Example 1 · Slope and tangent line

Find the slope of the tangent line to $f(x) = x^2 + 1$ at $x = 3$, and write its equation.

Solution. $\dfrac{(3+h)^2+1-10}{h} = \dfrac{6h+h^2}{h} = 6 + h \to 6$. The point is $(3, 10)$.

Slope 6; tangent line $y - 10 = 6(x-3)$.

Example 2 · A reciprocal

Find the instantaneous rate of change of $f(x) = \dfrac{4}{x}$ at $x = 2$.

Solution. $\dfrac{\frac{4}{2+h} - 2}{h} = \dfrac{4 - 2(2+h)}{h(2+h)} = \dfrac{-2}{2+h} \to -1$.

The rate of change is -1.

Example 3 · Velocity

A ball has height $H(t) = 50t - 5t^2$ metres after t seconds. Find its velocity at $t = 2$.

Solution. $\dfrac{H(2+k) - H(2)}{k} = \dfrac{50k - 20k - 5k^2}{k} = 30 - 5k \to 30$.

At $t = 2$ the ball is rising at 30 metres per second.

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Practice

CORE

1. Find the average rate of change of $f(x) = x^2 - 4$ on $[1, 3]$.

2. Find the slope of the tangent line to $f(x) = 3x^2$ at $x = 1$.

3. Write an equation of the tangent line to $f(x) = x^2 - 2x$ at $x = 4$.

4. Explain why the tangent line to $f(x) = 5x - 7$ has slope 5 at every point.

EXAM LEVEL

5. Find the slope of the tangent line to $f(x) = \dfrac{2}{x - 1}$ at $x = 3$.

6. Find the slope of the tangent line to $f(x) = \sqrt{x + 5}$ at $x = 4$.

7. Find the slope of the tangent line to $f(x) = x^3$ at $x = 2$, and write its equation.

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8. At what value of x does the graph of $f(x) = x^2 - 6x$ have a horizontal tangent line? Use a limit to find the slope at a general point x.

Question 9 · Free-response practice

A cart moves along a track so that its position after t seconds is $s(t) = t^2 + 4t$ metres.

(a) Find the average velocity of the cart on the interval $[1, 3]$.

(b) For $h \neq 0$, simplify the average velocity on $[2, 2+h]$, and use it to find the velocity at $t = 2$.

(c) Compare your answers to parts (a) and (b). Is this a coincidence? Explain using your expression from part (b).

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Worked solutions

1. $\dfrac{5-(-3)}{2} = 4.$

2. $\dfrac{3(1+h)^2-3}{h} = 6+3h \to 6.$

3. $\dfrac{(4+h)^2-2(4+h)-8}{h} = 6+h \to 6$; the point is $(4,8)$: $y-8=6(x-4)$.

4. $\dfrac{5(a+h)-7-(5a-7)}{h} = 5$ for every a and every h: a line is its own tangent.

5. $\dfrac{\frac{2}{2+h}-1}{h} = \dfrac{-1}{2+h} \to -\dfrac{1}{2}.$

6. $\dfrac{\sqrt{9+h}-3}{h} = \dfrac{1}{\sqrt{9+h}+3} \to \dfrac{1}{6}.$

7. $\dfrac{(2+h)^3-8}{h} = 12+6h+h^2 \to 12$; tangent line $y-8=12(x-2)$.

8. $\dfrac{(x+h)^2-6(x+h)-x^2+6x}{h} = 2x+h-6 \to 2x-6$, which is zero at $x=3$.

9. (a) $\dfrac{s(3)-s(1)}{2} = \dfrac{21-5}{2} = 8$ metres per second.

   (b) $\dfrac{(2+h)^2+4(2+h)-12}{h} = \dfrac{8h+h^2}{h} = 8+h$, which approaches 8 metres per second.

   (c) They are equal, and not by coincidence. On $[2-k, 2+k]$ the average velocity is $\dfrac{s(2+k)-s(2-k)}{2k} = \dfrac{16k}{2k} = 8$: for a quadratic, the average velocity over an interval equals the velocity at its midpoint.

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